

New Approaches to Energy Hardware Innovation Compared to Diffusion of Innovation Theory

Rogers' Diffusion of Innovation (DOI) Theory describes how new technologies spread through society, moving from innovators to early adopters and eventually the majority. While useful for consumer products, this framework assumes that technologies are already viable when they enter the market. For energy hardware and other "deep tech," the challenge often lies in bridging the technical and financial "Valleys of Death" before adoption can even begin.

The 2019 NREL white paper highlights how new models address this gap. Programs such as Wells Fargo IN2, Cyclotron Road, and the Lab-Embedded Entrepreneurship Program (LEEP) provide startups with non-dilutive funding, lab access, and trusted validation to reduce risk and ensure reward. Others, like Breakthrough Energy Ventures and The Engine by MIT, offer long-term equity funding designed to support high-cost, high-risk technologies. These approaches differ from DOI by focusing on de-risking and infrastructure rather than social adoption alone, which I felt to be very interesting but it makes a lot of sense for this market.

Another important aspect is how these newer models encourage collaboration between industries, investors, and researchers instead of keeping them in separate lanes. For example, when banks, labs, and entrepreneurs all work together from the start, the feedback loop is much faster and streamlined. Engineers can get real-world input on design needs, investors can see early signs of viability, and companies can adapt before wasting time and resources, something so many executives still struggle to get through. This kind of cross-disciplinary teamwork isn't really part of Rogers' DOI framework, but it's critical for hardware-heavy climate technologies that can't afford years of trial and error on their own.

From a student engineering perspective, what stands out is how these programs create a kind of support system that DOI doesn't really cover. Instead of just waiting for people to "adopt" an idea, they give startups the tools, labs, and money they need to prove their designs actually work in the real world. It feels less like a theory and more like hands-on problem solving; taking big risks early on so the technology can get a chance to get to the market. For

climate and energy systems, that extra support can make the difference between a great idea dying in the lab or becoming something that actually helps with decarbonization and a cleaner, bright future. As engineers, this kind of support system should be something that everyone takes advantage of, because of how influential and powerful the help can be.